

Barcode clustering benchmark

Preserved v14 + faster startup | 60 independent libraries per simulated design | six configurations

FN and FP: mean counts. F1: mean % (SD). Time: median workflow seconds. Milo: one fixed reference.

Method	FN (lower)	FP (lower)	F1 % (higher)	Time s (lower)
Random N20: substitutions + repeat-dependent indels				
barbac Hamming	124.92	18.30	99.28004 (0.0758)	1.40
barbac LV + Poisson	124.90	18.25	99.28038 (0.0752)	1.76
Shepherd [55/60]	n/a	n/a	n/a	n/a
Starcode sphere	129.72	22.32	99.23569 (0.0797)	3.29
Starcode MP	118.18	142.98	98.69582 (0.1112)	3.22
Bartender	125.17	48.03	99.13063 (0.0921)	1.89
Anchored N20 + AA/TT: substitutions + repeat-dependent indels				
barbac Hamming	127.70	24.90	99.23302 (0.0736)	1.52
barbac LV + Poisson	127.67	24.08	99.23726 (0.0741)	2.05
Shepherd [55/60]	n/a	n/a	n/a	n/a
Starcode sphere	132.63	27.65	99.19431 (0.0829)	3.77
Starcode MP	119.95	191.83	98.44674 (0.1384)	3.67
Bartender	127.73	137.97	98.67218 (0.0786)	2.31
Milo / Johnson published reference simulation				
barbac Hamming	469	87	99.72147	17.50
barbac LV + Poisson	471	83	99.72246	33.65
Shepherd	470	85	99.72196	120.12
Starcode sphere	771	353	99.43682	170.68
Starcode MP	532	560	99.45408	193.39
Bartender	496	795	99.35546	41.77

Bold/shading identifies numerical leaders among complete rows, not significance. 10 failed tool calls are retained. An incomplete row shows successes/planned in brackets; n/a means the full-design result is unavailable. Conditional summaries and all eight accuracy contrasts appear in the supplement.

FN: true identities absent from inferred centroids. FP: inferred identities absent from truth. Exact-identity F1 includes zero-read true barcodes. Read assignment, positive-read truth and abundance metrics are reported separately.

New simulations: 60 independent seeds, 10,000 true identities and one million expected reads per library; paired designs share variable identities and parent counts. Substitutions: 0.4% per base. Indels: unscaled archived homopolymer-specific rates.

The calibrated table ends at repeat length 13. Further indel recursion is stopped for affected reads beyond that support; reads, substitutions and truth are retained. Boundary use: 249 reads across 16 of 120 libraries. This is a declared finite-support model, not an empirical estimate beyond the table.

Milo: unchanged deposited simulation, 100,000 true barcodes and 24,996,128 reads; used during development. Its single-reference scores do not establish independent-library statistical superiority. Its timing is one observation in this final campaign.

All methods use distance 3 and one thread. Barbac: support ordering, ratio 20, configured error proxy 0.005, design scoring off, LV Poisson on. Hamming retains rare-indel rescue. Shepherd: nominal length, Bayes threshold 4. Starcode MP ratio 5. Bartender: seed length 5, step 1, z=5, cutoff 1.

Timing: serial fresh-worker elapsed time including startup, required conversion and centroid/member exports; staging, scoring and hashing excluded. The 28 calls overlapping a system sleep/partial-wake interval were repeated once with sleep prevented; mappings had to match exactly. Original and repaired records are retained. Other cells ran once.

The publication scope was selected after development inspection. Code, settings, seeds and test methods were frozen before final generation. A disclosed reporting addendum retains tool failures and tests only fully observed 60-pair contrasts, keeping the eight-comparison correction. All registered libraries remain. Development controls are preserved.